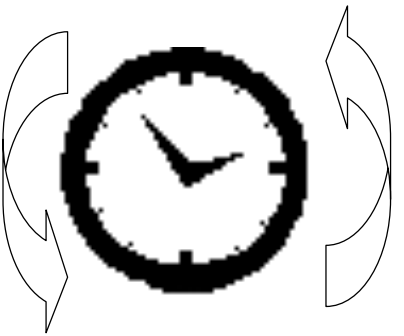
Solar means sun. The solar system has various parts: the sun, 8 official planets, their moons, asteroids (small rocky bodies), comets (small icy objects) several dwarf planets; such as Pluto and, mostly, space.

The planets orbit around the sun. An orbit means the path an object in space makes. Orbits are usually counterclockwise.



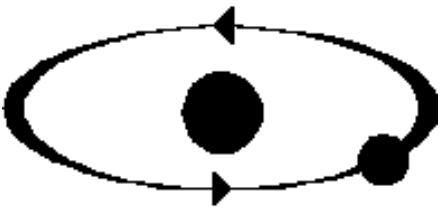
The orbits of planets, their moons, asteroids, and comets in the solar system are elliptical, or oval shaped.

trace the dotted words.

An elliptical orbit has 2 foci, or 2 focus points. One focus is the sun and the other point is out in space where the sun's gravitational pull tugs the object back instead of the object continuing out of the solar system by momentum.

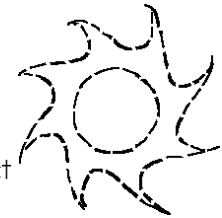
To rotate means to spin.

To revolve means to go around another object.



trace the dotted words.

Solar → means the sun.



Orbit → means the path an object takes around another object.

Orbits → are elliptical, oval-shaped.

Revolve → means one object moves around another.

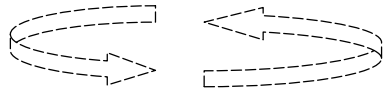
Rotate → means to spin.

Momentum → means an object keeps moving.

Gravity → is a force, a pull.

Space → is total emptiness, nothing

Counterclockwise



dotted words and pictures.

revolve

orbit elliptical gravity rotate

foci space sun

1. Total emptiness, nothing, is called s _____.

2. The sun and the planets each spin or ro _____.

3. The planets, their moons, asteroids and comets, all

ro _____ around the sun.

4. The force that is the pull of the earth on an object is called

g _____.

5. The path an object takes around another is called an

o _____.

6. The sun is one of the f _____ that a planet swings around.

7. Solar means s _____.

8. Most orbits are e _____, an oval shape.

Fill in the blanks and trace the dotted letters.

revolve

orbit elliptical gravity rotate

foci space sun

1. Total emptiness, nothing, is called _____.
space

2. The sun and the planets each spin or _____.
rotate

3. The planets, their moons, asteroids and comets, all

_____ around the sun.
revolve

4. The force that is the pull of the earth on an object is called

_____.
gravity

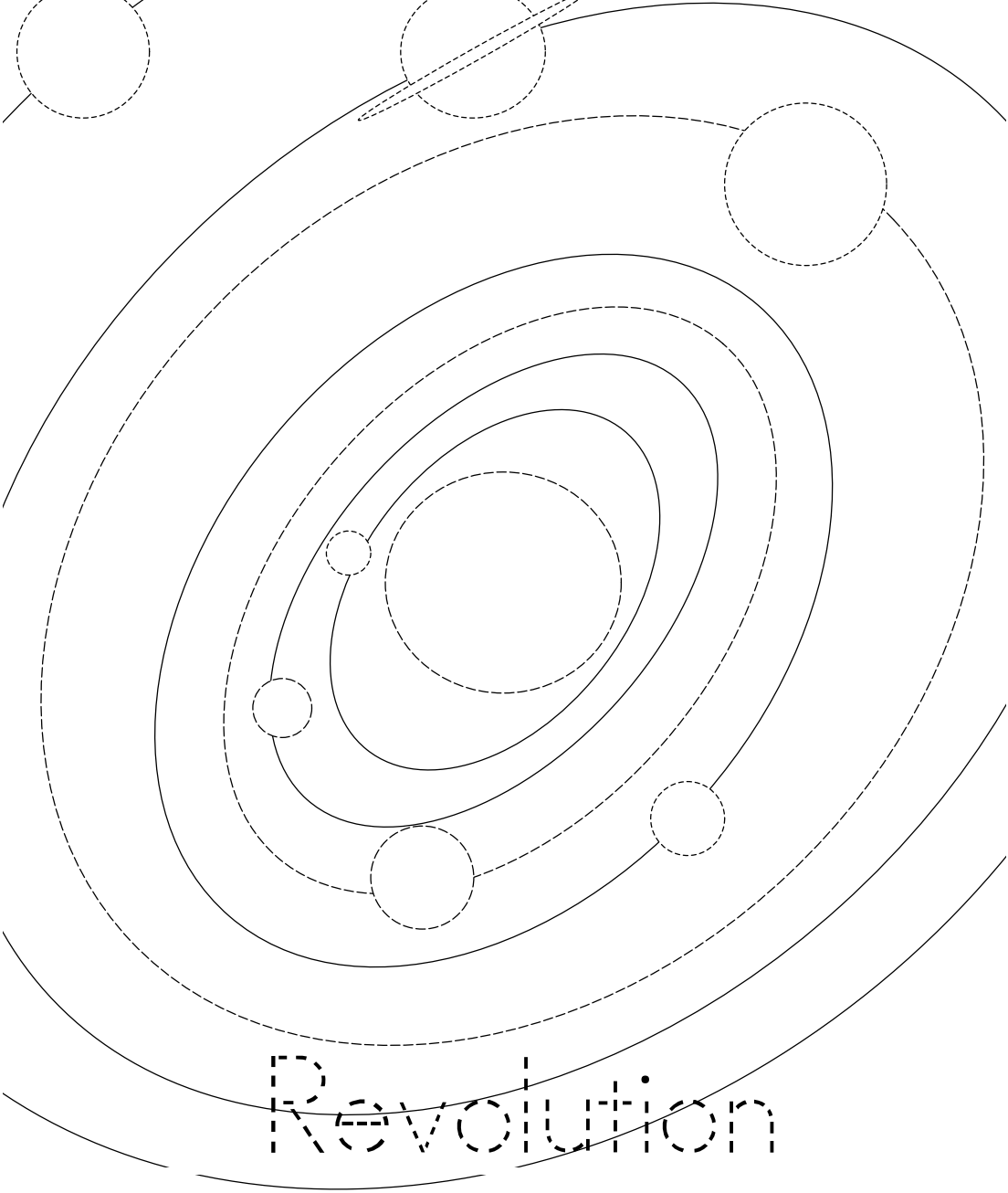
5. The path an object takes around another is called an

_____.
orbit

6. The sun is one of the _____ that a planet swings around.
foci

7. Solar means _____.
sun

8. Most orbits are _____, an oval shape.
elliptical



Trace the dotted lines to complete the pictures.

Our sun is a middle-aged star called Sol.

Our sun is a yellow star.

Our sun is one of the foci around which the planets, asteroids, and comets revolve in elliptical orbits.

Our sun has eight official planets revolving around it with an asteroid belt between the inner and outer planets as well as comets.

Our sun rotates and has solar storms and solar flares.

Our sun provides virtually all the energy on our planet. Our sun provides the light energy for photosynthesis (to make food with light as the energy source) in plants on planet Earth.

Nuclear fusion occurs in the sun, combining two atoms of the element hydrogen into one atom of helium.

Read the information and trace the dotted words.

Word Bank

photosynthesis Sol
yellow energy rotates
nuclear fusion planets
gravity oval
momentum

1. Our sun's name is S_____.

2. The sun is a Y_____, middle-aged star.

3. The sun RO_____ and has solar storms and solar flares.

4. The P_____ revolve around the sun in elliptical orbits.

5. An elliptical orbit has the shape of an O_____.

6. M_____ means an object moving wants to continue moving.

7. G_____ is a force, a pull of the earth on an object.

8. The sun gives off light and heat EN_____.

9. The plants on our planet use the light energy in

P_____ to make food, glucose.

10. N_____ F_____ is the combining of two atoms of hydrogen into one atom of another element, helium.

Use the words in the Word Bank to fill in the blanks.

The name of our sun is	yellow
Light energy from the sun is used by plants to make food.	planets
	Sol
Two hydrogen atoms are joined to form one atom of a new element, helium.	light energy
The sun is orbited by 8 official	nuclear fusion
The sun is not an old star, or a young star, it is	middle-aged
The sun gives off	
Our sun is this color	photosynthesis

Match the definition with the correct dotted word.

The name of our sun is

Light energy from the sun is used by plants to make food.

Two hydrogen atoms are joined to form one atom of a new element, helium.

The sun is orbited by 8 official

The sun is not an old star, or a young star, it is

The sun gives off

Our sun is this color

yellow

planets

Sol

light energy

nuclear fusion

middle-aged

photosynthesis

P	A	P	P	L	E	H	C	A	T	B	A	N	D	M	E	T	S	Y	S
H	Y	D	R	O	G	E	N	E	L	D	O	O	D	E	E	K	N	A	Y
O	V	A	L	P	E	L	L	I	P	T	I	C	A	L	N	O	O	K	N
T	P	L	A	C	E	I	L	L	U	S	I	O	N	I	N	N	E	R	T
O	D	O	N	E	P	U	L	L	G	R	E	H	M	R	W	W	R	G	H
S	O	L	J	S	E	M	O	M	E	N	T	U	M	K	K	T	Y	P	E
Y	E	L	L	O	W	J	L	S	R	Y	R	E	N	B	D	G	R	L	T
N	U	C	L	E	A	R	F	U	S	I	O	N	R	W	H	F	G	A	I
T	S	F	H	L	H	R	O	T	A	T	E	E	J	S	F	E	H	N	C
H	E	G	J	L	E	G	C	H	S	A	E	R	E	V	O	L	V	E	E
E	H	E	F	I	J	N	I	W	E	F	H	G	R	J	R	R	N	T	A
S	R	G	V	P	S	G	R	A	V	I	T	Y	W	E	C	T	H	S	R
I	B	S	H	T	B	D	M	T	D	N	G	R	T	N	E	V	S	E	T
S	D	E	F	I	N	R	W	N	F	Y	K	E	W	N	Y	P	P	A	H

Word Bank

hydrogen	helium	yellow	energy
photosynthesis	rotate	revolve	foci
planets	gravity	pull	force
momentum	elliptical	oval	sol
	nuclear fusion		

Find and circle the words from the Word Bank.

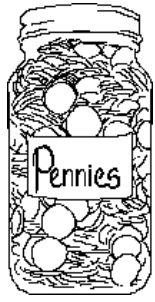
The inner planets around the sun are Mercury, Venus, Earth, and Mars.

The word "planet" means "wanderer". Planets are big bodies orbiting the sun. The inner planets around the sun, Mercury, Venus, Earth, and Mars, are terrestrial or rocky planets. The inner planets around the sun have no rings, few satellites, and a slow rotation (spin), and high densities.

Satellites that are natural satellites are called moons. Artificial satellites are man made and sent into space.

Density refers to the mass (amount of matter) in a certain amount of space. A jar of pennies would have more density than the same size jar filled with feathers.

Low Density

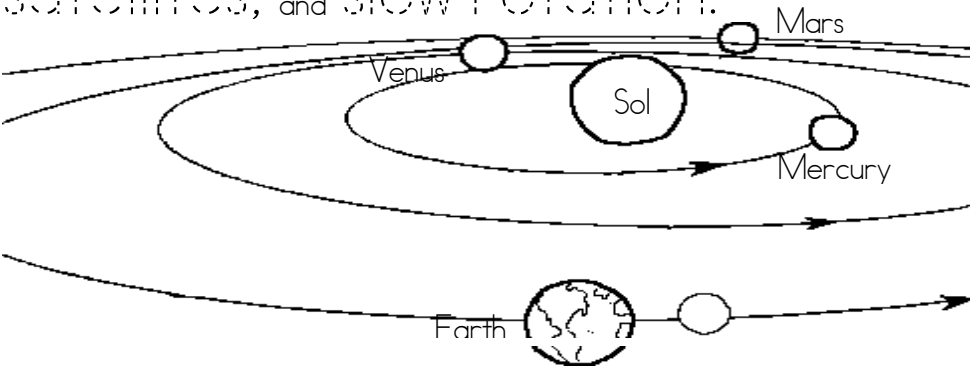


High Density

The Inner Planets

Read the information and trace the dotted words.

Planets are big bodies orbiting the sun. The word "planet" means "wanderer". The inner planets that orbit the sun are: Mercury, Venus, Earth, and Mars. The inner planets rotate slowly and revolve around the sun. Some planets have natural satellites. A natural satellite is like a moon. Our inner planet, Earth, has one natural satellite. The inner planets, Mercury, Venus, Earth, and Mars, are terrestrial (rocky). The inner planets have a solid surface, high density, no rings, few satellites, and slow rotation.



The Inner Planets

Read the information and trace the dotted words.

1. _____ are big bodies that orbit the sun.
   Moons Planets Stars

2. _____ is one of the inner planets.
   Neptune Uranus Earth

3. The inner planets _____ _____ and revolve around the sun.
   do **not** rotate rotate **slowly** rotate **quickly**

4. **Inner** planets are _____.
   terrestrial gaseous

5. The inner planets have ____ _____.
   many satellites no rings fast rotation

6. The planets with **high density** are the _____ _____.
   space outer planets inner planets

7. A **natural satellite** is like a _____.
   planet star moon

8. The word planet means _____.
   Sol "wanderer" Luna

9. The inner planets revolve around the _____.
   sun earth comets

Circle the correct answer to fill in the blanks.

The Inner Planets

1. _____ are big bodies that orbit the sun.
   Moons Planets Stars

2. _____ is one of the inner planets.
   Neptune Uranus Earth

3. The inner planets _____ _____ and revolve around the sun.
   do **not** rotate rotate **slowly** rotate **quickly**

4. **Inner** planets are _____.
   terrestrial gaseous

5. The inner planets have _____.
   many satellites no rings fast rotation

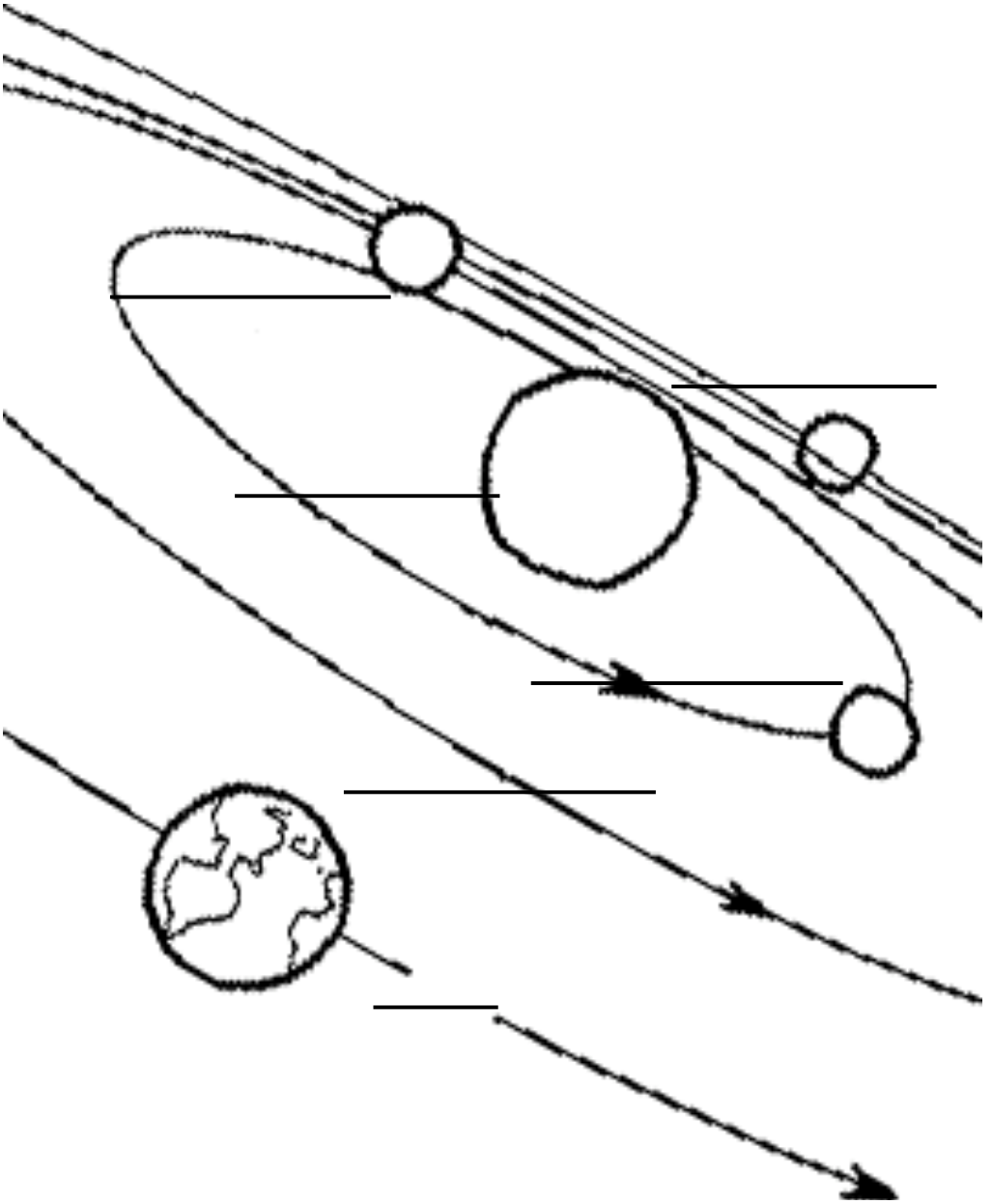
6. The planets with **high density** are the _____ _____.
   space outer planets inner planets

7. A **natural satellite** is like a _____.
   planet star moon

8. The word planet means _____.
   Sol "wanderer" Luna

9. The inner planets revolve around the _____.
   sun earth comets

Planets



The Inner Planets

Label the sun and each of the inner planets. Draw Earth's moon.

Earth is one of the inner planets, the third planet from the sun. Earth is also called Terra.

Earth is the inner planet on which you live. Earth has a solid surface, no rings, a high density, and one natural satellite, the moon. Earth has a slow rotation. Earth rotates (spins) slowly, once every 24 hours, causing night and day. Earth has a revolution around the sun that takes 365 days, or a year. Earth has an atmosphere (layers of air), water, and living things (plants and animals). Earth has one natural satellite, the moon, which affects the water on the earth and causes tides. The moon is called Luna. The moon reflects the light from the sun. The different amounts of light reflected to earth from the sun are called phases of the moon.

Planet Earth and its Moon

Read the information and trace the dotted words.

1. The Earth is one of the inner planets.

2. The Earth has an atmosphere, water, and living things, such as plants and animals.

3. The Earth has a solid surface, no rings, and a high density.

4. Earth rotates every 24 hours which is called a day.

5. Earth revolves around the sun in an elliptical orbit once every 365 days, which is a year.

6. Earth has one natural satellite, the moon, which has craters and is named Luna.

7. The moon reflects light from the sun like a mirror reflects light.

8. The different amounts of light as you see it from Earth are called phases.

9. The moon affects Earth's water and causes tides.

Planet Earth and its Moon

Read the information and trace the dotted words.

Word Bank

365 days inner moon

craters no water solid atmosphere

living 24 hours

1. The Earth is one of the _____ planets.

2. The Earth has a _____ surface and is called terrestrial.

3. The Earth rotates every _____ _____, causing day and night.

4. The Earth revolves every _____ _____, called a year.

5. The moon has no atmosphere and meteorites strike it causing

   _____.

6. The Earth has _____ things, like plants and animals.

7. The Earth is an inner planet and has _____ rings.

8. The word Luna refers to the _____.

9. The Earth has _____, frozen at the poles and liquid at the equator.

10. The earth has layers of air or an _____.

Fill in the blanks with the words from the Word Bank.

Word Bank

365 days inner moon

craters no water solid atmosphere

living 24 hours

1. The Earth is one of the _____ planets.
   inner
2. The Earth has a _____ surface and is called terrestrial.
   solid
3. The Earth rotates every _____ _____, causing day and night.
   24 hours
4. The Earth revolves every _____ _____, called a year.
   365 days
5. The moon has no atmosphere and meteorites strike it causing _____.
   craters
6. The Earth has _____ things, like plants and animals.
   living
7. The Earth is an inner planet and has _____ rings.
   no
8. The word Luna refers to the _____.
   moon
9. The Earth has _____, frozen at the poles and liquid at the equator.
   water
10. The Earth has layers of air or an _____.
    atmosphere

Planet Earth and its Moon

Word Bank

rotate	revolve	inner planet	natural satellite
water	Luna	Terra	no rings
solid surface	craters	high density	moon
atmosphere	year	day	reflects
phases		tides	

A	R	E	F	I	E	C	T	S	F	H	D	R	F	J	H
T	E	C	R	A	T	E	R	S	Y	H	T	O	H	T	I
M	K	T	Z	F	E	E	W	H	E	M	T	T	W	J	G
O	N	T	F	D	R	S	Y	H	A	N	S	A	W	U	H
S	J	I	S	G	R	M	R	J	R	N	R	T	R	M	D
P	N	D	L	D	A	Z	T	T	D	N	T	E	W	L	E
H	L	E	D	S	R	E	V	O	L	V	E	R	A	R	N
E	U	S	T	R	J	K	U	D	R	Y	E	F	T	J	S
R	N	O	R	I	N	G	S	J	F	D	E	T	E	K	I
E	A	T	J	J	T	E	D	E	D	A	Y	H	R	H	T
O	F	G	F	E	P	H	A	S	E	S	E	E	T	I	Y
N	A	T	U	R	A	L	S	A	T	E	L	L	I	T	E
E	J	S	O	L	I	D	S	U	R	F	A	C	E	X	Y
I	N	N	E	R	P	L	A	N	E	T	Q	M	O	O	N

Find and circle all the words from the Word Bank.

Planet Earth and its Moon

The outer planets, Jupiter, Saturn, Uranus, and Neptune, are huge and called Jovian and are gas planets.

The outer planets, Jupiter, Saturn, Uranus, and Neptune, have no solid surface; they are gas planets having the elements hydrogen and helium gases.

The outer planets, Jupiter, Saturn, Uranus, and Neptune, have low densities (like a jar of feathers) and rapid rotations (spins). The outer planets, Jupiter, Saturn, Uranus, and Neptune have rings around them.

The outer planets, Jupiter, Saturn, Uranus, and Neptune, have lots of satellites. Between the inner and outer planets is an asteroid belt.

The Outer Planets

Read the information and trace the dotted words.

The outer planets are: Jupiter Saturn, Uranus, and Neptune.

An asteroid belt separates the outer planets from the inner planets.

The outer planets have low densities, less mass in the space the planet occupies, like the low density of a jar of cotton balls compared to a jar of rocks.

rocks

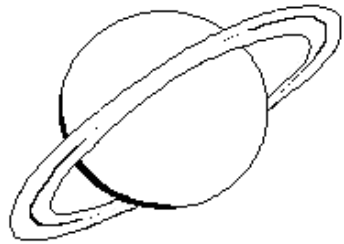
cotton balls

Jupiter Saturn, Uranus, and Neptune are called Jovian because they are big, gaseous, and have no solid surface. Each planet has rings and many satellites.

The Outer Planets

Read the information and trace the dotted words.

1. The outer planets like _____ have rings.
   Mars Saturn Venus



2. The outer planets include _____.
   Neptune Earth Mercury

3. The outer planets have a _____ density.
   high low no

4. The outer planets rotate or spin _____.
   slowly rapidly

5. The outer planets have many _____ satellites.
   natural artificial no

6. The outer planets are planets with no _____ surface.
   solid gaseous

7. Between the inner and outer planets is an _____ belt.
   comet moon asteroid

8. The most abundant gases in the outer planets are

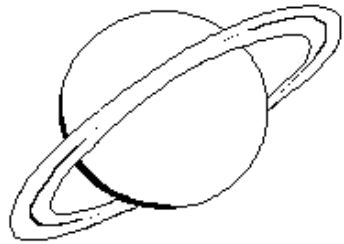
   _____.
   oxygen/carbon dioxide nitrogen/neon hydrogen/helium

9. The outer planets are called _____ because they are so huge.
   dwarf terrestrial Jovian

The Outer Planets

Circle the correct answer to fill in the blank.

1. The outer planets like _____ have rings.
   Mars Saturn Venus



2. The outer planets include _____.
   Neptune Earth Mercury

3. The outer planets have a _____ density.
   high low no

4. The outer planets rotate or spin _____.
   slowly rapidly

5. The outer planets have many _____ satellites.
   natural artificial no

6. The outer planets are planets with no _____ surface.
   solid gaseous

7. Between the inner and outer planets is an _____ belt.
   comet moon asteroid

8. The most abundant gases in the outer planets are

   _____.
   oxygen/carbon dioxide nitrogen/neon hydrogen/helium

9. The outer planets are called _____ because they are so huge.
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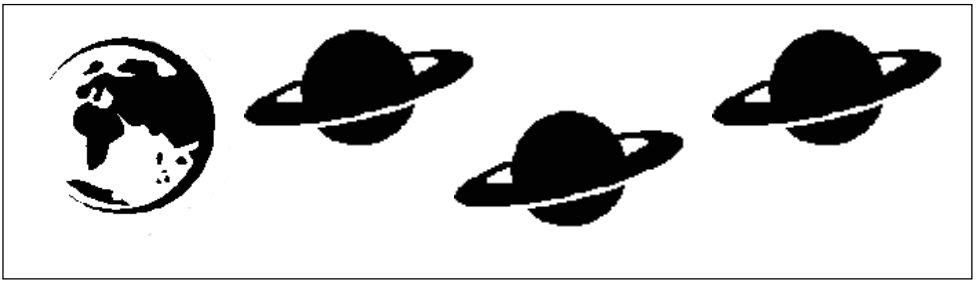
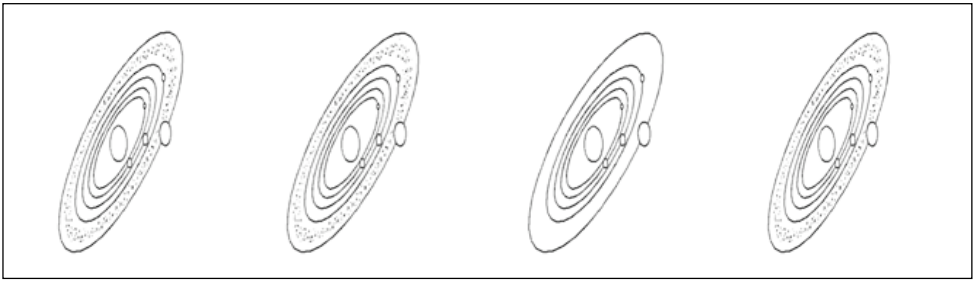
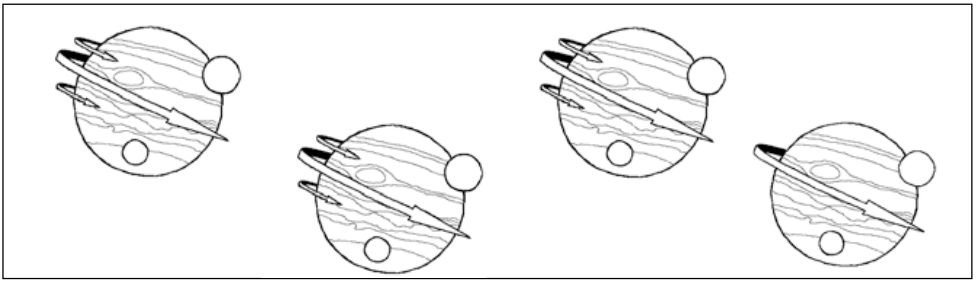
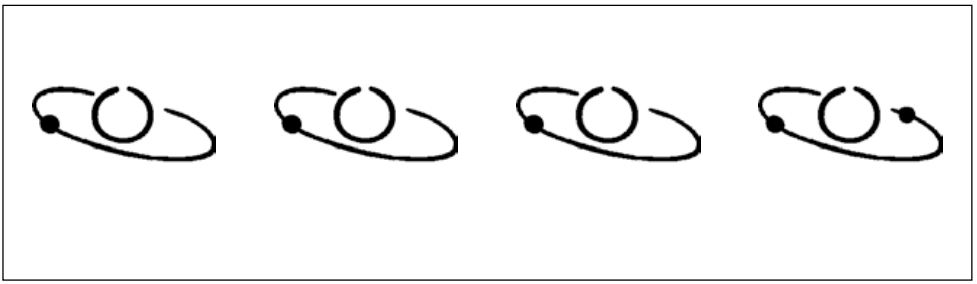
The Outer Planets





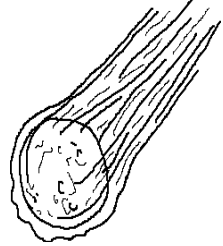




The Outer Planets

Circle the picture in each row that is different.

A comet is a small icy body with a highly erratic elliptical orbit. A comet is composed of a head and a tail. The tail of a comet is always pointing away from the sun.



Asteroids are small rocky bodies traveling through space. An asteroid belt is a collection of small rocky bodies in a cluster revolving around the sun. There is an asteroid belt in our solar system between the inner and the outer planets.

Satellites are small bodies revolving around another larger body. A natural satellite is like our moon, which revolves around the Earth. An artificial satellite is man-made. There are artificial satellites for communication and weather revolving around the Earth.

A meteor is a tiny rocky body. If meteors do not burn up in the atmosphere, their impact creates a crater if the planet or moon has a solid surface. Planets or moons without an atmosphere have many craters.

Read the information and trace the dotted words.

Satellites are small bodies that revolve around a larger body like our moon revolves around the Earth. A moon is a natural satellite. Man-made satellites are called artificial satellites and are used for weather and communication. A comet is an icy body with a head and a tail that always points away from the sun. Comets have very erratic elliptical orbits. Meteors are tiny rock bodies that are called meteorites when they strike the ground and create a crater. Meteors can be small enough to burn up in the atmosphere of a planet. A planet or moon without an atmosphere has lots of craters.

Asteroids are small rocky bodies. When asteroids are in a group revolving around the sun, they can form an asteroid belt like the one between the inner and outer planets.

Comets, and Meteorites

Read the information and trace the dotted words.

An icy body with a head and a tail •

A natural satellite •

Found between the inner and outer planets •

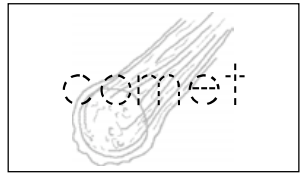
A meteor striking a solid surface causes a •

Large rocky body •

Man-made weather satellite •

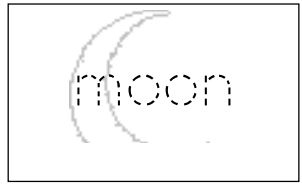
•

• asteroid

• 

• asteroid belt

• artificial satellite

• 

Comets, and Meteorites

Trace dotted words, match definitions to words and pictures.

An icy body with a head and a tail

A natural satellite

Found between the inner and outer planets

A meteor striking a solid surface causes a

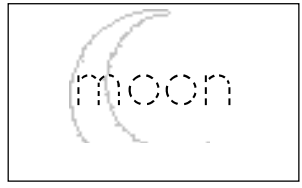
Large rocky body

Man-made weather satellite

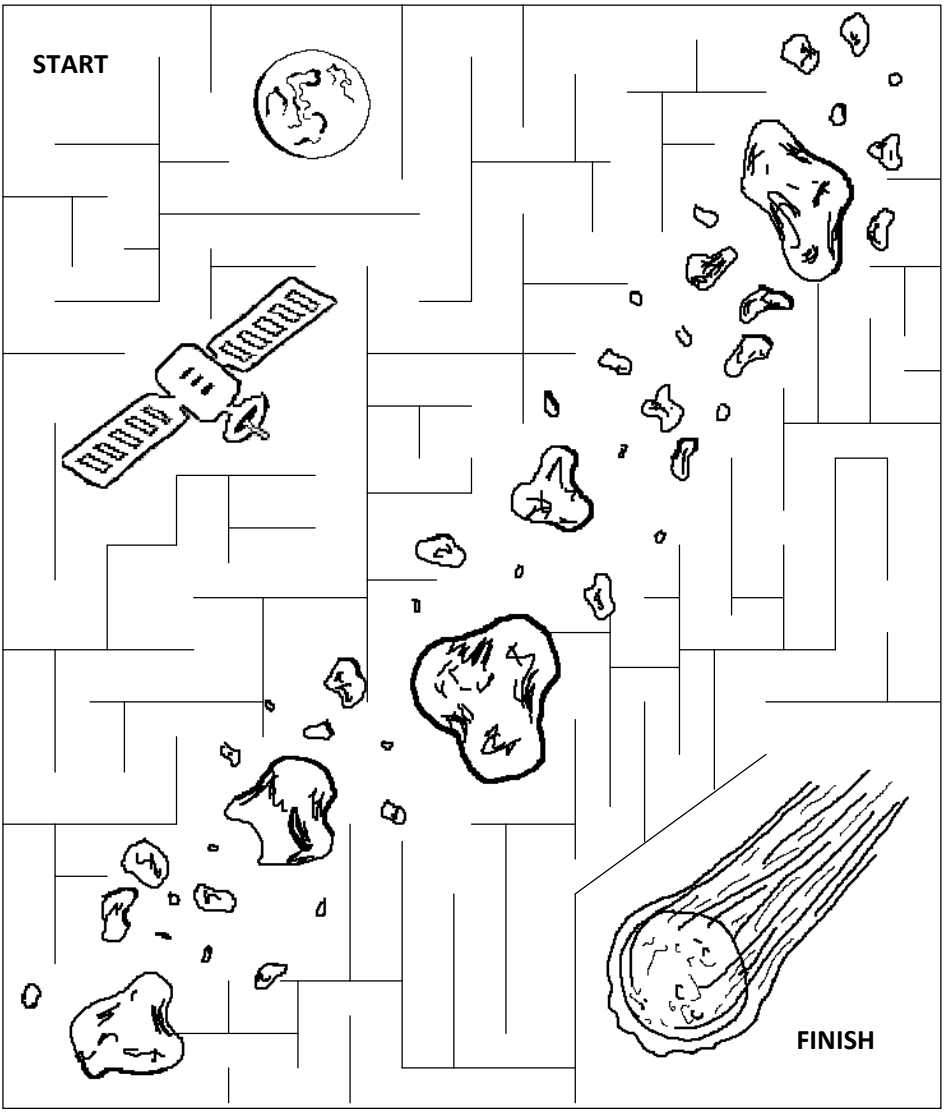
asteroid

asteroid belt

artificial satellite



Comets, and Meteorites



Comets, and Meteorites

Trace a path to the comet at the finish of the maze.

A	S	T	E	R	O	I	D	B	E	L	T	R	H
G	N	E	P	T	U	N	E	J	S	T	A	R	Y
W	U	H	E	S	E	J	N	T	A	M	I	E	D
A	C	B	C	R	A	T	E	R	T	E	L	V	R
N	L	S	E	R	R	H	R	R	U	G	E	O	O
D	E	N	S	I	T	Y	G	H	R	R	W	L	G
E	A	C	P	N	H	M	Y	R	N	A	R	V	E
R	R	O	L	G	U	E	P	V	R	V	J	E	N
E	F	M	A	S	R	R	H	E	L	I	U	M	Y
R	U	E	N	E	A	C	O	N	S	T	P	E	E
S	S	T	E	E	N	U	T	U	R	Y	I	H	L
H	I	E	T	W	U	R	O	S	C	X	T	Z	L
S	O	L	A	R	S	Y	S	T	E	M	E	M	O
Z	N	X	T	C	S	D	Y	B	N	E	R	E	W
A	N	I	M	A	L	S	N	R	O	T	A	T	E
G	P	N	O	F	U	G	T	M	Y	E	W	E	G
A	L	B	S	F	N	H	H	M	O	O	N	O	R
S	A	D	P	J	A	S	E	R	E	R	T	R	M
E	N	A	H	K	T	S	S	R	J	I	Y	R	H
O	T	Y	E	A	R	R	I	T	J	T	S	K	E
U	S	G	R	M	A	R	S	T	T	E	R	R	A
S	A	T	E	L	L	I	T	E	M	S	O	L	D

solar system
asteroid belt
energy
rings
Venus
Uranus
Terra
moon
satellite
photosynthesis
comet
Earth
Neptune
yellow
crater
gaseous
atmosphere
revolve
Saturn
planet
star
meteor
helium
solid surface
rotate
Jupiter
Sol
gravity
meteorites
hydrogen
density
Mars
Mercury
Luna

Find and circle all the words from the Word Bank.